

$$I_{rms} = 2I \quad \Rightarrow I_o = 2\sqrt{2}I$$

23 B S experiences attractive forces towards P and R (Like currents) but since R has a higher current, net force is towards R. Q has a current in the opposite direction so S experiences another force away from Q. Resultant force is downwards.

24 A $F_B = N \times (BIL \sin\theta) = 100 \times 0.55 \times 1.6 \times 0.1$